

Program : Diploma in Printing Technology	
Course Code : 6101A	Course Title: Specialized Printing Applications
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To provide knowledge about the specialized applications of printing processes in the changing world and the infinite opportunities in the sector.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Gravure, Flexographic printing process		Mechanism of printing machines II	3
Screen printing		Mechanism of printing machines II	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain Security features of printed products.	15	Understanding
CO2	Explain 3D printing and applications.	14	Understanding
CO3	Explain Lenticular and hybrid printing.	15	Understanding
CO4	Explain Textile and printed electronics.	14	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain Security features of printed products.		
M1.01	General security features	3	Understanding
M1.02	Classification of ink	5	Understanding
M1.03	Classification of paper	4	Understanding
M1.04	Security products	3	Understanding
Contents: Security printing - Definition. Goals of security printing. General Security features- water mark, security thread, latent image, micro lettering, Intaglio Printing, identification mark, see through register, hologram, Pantograph screens, Void pantograph Screen, ODT - optical deterrent technology, Guilloches, Warning bands, Code safe, High resolution graphics and Padlock icon. Various printing methods used for security printing. Basic classification of security inks - Trademark colors, Color changing ink, Magnetic ink, Copy protection ink, Erasable ink, Fugitive ink, Pen reactive ink, Heat reactive ink, Coin reactive inks, Migrating ink, Bleeding inks, Florescent ink, Metallic ink and UV ink etc. Basic principles of security papers - Safety paper, Chemical reactive paper, Special papers, Water mark paper and Copy evident paper etc. Types and usages of security threads - Metalized thread, Windowed thread, Holographic windowed thread. Micro text, Clear text and Thermo text. Security features of currency, cheque, cheque layout, credit card, passport manufacturing and features.			
CO2	Explain 3D printing and applications.		
M2.01	Introduction to 3D printing	4	Understanding
M2.02	Printing process	4	Understanding

M2.03	Applications	3	Understanding
M2.04	Nanography	3	Understanding
	Series Test I	1	
Contents: 3D printing: Introduction about 3D printing. 3D printing process principle. 3D printing technologies - Extrusion deposition, Granular materials binding, Laminated, Photo-polymerization. Steps involved in 3D printing process. Application of 3D printing. Introduction of (brief) - Bio printing. Nano scale 3D printing. Nanography: definition, inventor. Nanography process. Advantages and disadvantages of Nanography. Nano-printing machines and its features. Comparison between Nanography and offset.			
CO3	Explain Lenticular and hybrid printing.		
M3.01	Lenticular printing	6	Understanding
M3.02	Hologram	5	Understanding
M3.03	Hybrid printing	4	Understanding
Contents: Lenticular printing: Lenticular printing definition. Basic principles of Lenticular printing process. Construction of Lenticular image. Types of Lenticular prints. Principle of Lenticular lens. Lenticular Image and interlacing. Steps involved in Lenticular printing process. Effects of Lenticular print - flip, 3D, animation, zoom and morphing Lenticular. Hologram: Basic principles of holograms making process. Hologram types. Components of hologram making system - laser, lenses, beam splitter, mirrors, holographic film. Process steps of hologram making system. How hologram works. Requirements for producing hologram - object, laser (object beam, reference beam), recording medium, environment. Hybrid printing: definition of hybrid printing. Introduction to hybrid printing machines. Hybrid printing. DI presses. Names of manufacturers and machines for hybrid printing.			
CO4	Explain Textile and printed electronics.		
M4.01	Waterless offset	3	Understanding
M4.02	Dry offset	4	Understanding
M4.03	Textile printing	3	Understanding
M4.04	Printed Electronics	4	Understanding
	Series Test II	1	
Contents: Waterless offset: Basic principles of waterless offset printing. Plate structure of waterless offset printing. Merits and Demerits. Dry offset: Principle of dry offset. Dry offset process. Dry offset plates - types, manufacturing methods. Dry offset printing machines. Difference			

between waterless and dry offset printing. Print on demand: Print on demand - definition, application, advantages and disadvantages. Textile printing: Definition of textile printing. Textile printing steps. Methods of printing: Hand block printing, Perrotine printing, stencil, screen, digital, block printing, burn out printing, blotch, digital, duplex, spray printing, warp printing. Digital textile printing. Digital printing machines used for textile printing. Screen printing machines used for textile printing. Printed Electronics: Definition of printed electronics. Advantages of printed electronics. Conventional print processes used in printed electronics. Application of printed electronics.

Text / Reference:

T/R	Book Title/Author
T1	The art of paper currency , Quarlet books ltd, Martin Monestics.
R1	Barcodes – design, printing and quality control , William H Erdei.
R2	3D Printing 2nd edition , Christopher Barnatt.
R3	Hand book of print media , Helmut Kipphan.
R4	Printing in a digital world , David Bergsland.

Online resources:

Sl.No	Website Link
1	https://www.livescience.com
2	https://www.agfagraphics.com
3	https://www.livescience.com